

Effects of ginger on motion sickness and gastric slow-wave dysrhythmias induced by circular vection.

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Ginger has long been used as an alternative medication to prevent motion sickness. The mechanism of its action, however, is unknown. We hypothesize that ginger ameliorates the nausea associated with motion sickness by preventing the development of gastric dysrhythmias and the elevation of plasma vasopressin. Thirteen volunteers with a history of motion sickness underwent circular vection, during which nausea (scored 0-3, i.e., none to severe), electrogastrographic recordings, and plasma vasopressin levels were assessed with or without ginger pretreatment in a crossover-design, double-blind, randomized placebo-controlled study. Circular vection induced a maximal nausea score of 2.5 ± 0.2 and increased tachygastric activity and plasma vasopressin. Pretreatment with ginger (1,000 and 2,000 mg) reduced the nausea, tachygastria, and plasma vasopressin. Ginger also prolonged the latency before nausea onset and shortened the recovery time after vection cessation. Intravenous vasopressin infusion at 0.1 and 0.2 U/min induced nausea and increased bradygastric activity; ginger pretreatment (2,000 mg) affected neither. Ginger effectively reduces nausea, tachygastric activity, and vasopressin release induced by circular vection. In this manner, ginger may act as a novel agent in the prevention and treatment of motion sickness.